RATIONAL-DERIVED QUARTICS IN LEAN: A CORRECTED REPEATED-ROOT CLASSIFICATION AND THE BOUNDARY OF CONJECTURE 1

HIROKI FUKUI

ABSTRACT. A polynomial over $\mathbb{Q}$ is *rational-derived* if it and all of its derivatives split completely over $\mathbb{Q}$. Buchholz and MacDougall (J. Number Theory 81 (2000)) presented a classification of rational-derived quartics, their Theorem 7, whose four-distinct-root clause is entangled with their Conjecture 1. We report a three-part study of this classification, conducted through a Lean 4 formalization. First, an *audit*: the published proof of Theorem 7 contains a circularity and a quantifier gap — the argument assumes that the critical vertical translate is rational-derived, an assumption equivalent to the splitting it is required to prove (the equivalence is kernel-certified), and then concludes universally without discharging it; a single quartic, read over two base fields, refutes over $\mathbb{Q}$ the implication left once the circular assumption is removed, and over $\mathbb{Q}(\sqrt{3441})$ the statement's natural analogue under the full K-derived hypothesis. Second, a *repair*: Theorem 7′, a rank-free image theorem identifying the normalised $p(2,1,1)$ sector exactly with the image of the rational points of the elliptic curve $E\colon z^2 = w(w-6)(w+18)$ (Cremona `576i2`) under an explicit rational map — which, together with a kernel-verified affine normalisation and the elementary multiplicity cases, yields an unconditional classification of all repeated-root quartics; both directions reduce to explicit polynomial certificates together with elementary field and order arguments, and *no rank computation enters* — a proof found because the library lacked Mordell–Weil rank. The main statement compiles with axiom footprint exactly `[propext, Classical.choice, Quot.sound]`. Third, a *boundary*: a kernel-verified bridge identifies the normalised form of Conjecture 1 with the nonexistence of nondegenerate rational points on an explicit splitting surface; the three double covers governing the vertical-translate route are proved nontrivial over the splitting extension, with the kernel boundary of each claim marked in the text, and the fibre correspondence between the disjuncts of Theorem 7 and those covers is closed in the kernel. Conjecture 1 itself remains open: this development maps its boundary, it does not resolve it. The formalization detected the gap, sharpened the statements, and certified the repair; we record the mathematics together with the lessons of the formalization.

2020 *Mathematics Subject Classification.* Primary 68V20; Secondary 11G05, 11D41.

Key words and phrases. rational-derived polynomials, elliptic curves, formal verification, Lean, mathlib, proof auditing, proof repair.

1. Introduction

Around 1950, M. S. Klamkin exhibited a quartic polynomial, $(x-193)(x-141)(x+167)^2$, with the property that it and all of its derivatives have integer roots. For decades it stood as a curiosity: Carroll, writing in the Monthly in 1989, conjectured that it was essentially the only quartic of its kind with three or more distinct roots [6]; in the same year Richard Guy's problem column printed four pairs of roots of further examples, each pair larger than the last, the fourth running to twenty digits [9]. In 1995 Buchholz and Kelly organized the family by an elliptic curve [1], and in 2000 Buchholz and MacDougall presented a classification of all such *rational-derived* quartics, their Theorem 7, reducing what remained to a conjecture [2]. One outcome of the present study (Theorem 3.2) is that the scattered record has a single organizing point: Guy's four printed pairs are, in order and up to one root exchange, the images of the consecutive multiples $2P$, $3P$, $4P$, $5P$ of a single point P on a single elliptic curve of conductor 576 — with Klamkin's quartic the first of them. The examples, scattered over a seventy-six-year record, can be identified — across the different models used in the literature — with consecutive multiples of a single point on a single curve.

Call $f \in \mathbb{Q}[x]$ *rational-derived* if f and every derivative of f split completely over $\mathbb{Q}$; for a field $K \supseteq \mathbb{Q}$, call f *K-derived* if f and all its positive-degree derivatives split completely over K, and *properly* K-derived if it is K-derived but not rational-derived. Linear, quadratic and cubic rational-derived polynomials are completely understood; for quartics, sorting by root multiplicities, the types $p(4)$ and $p(3,1)$ are always rational-derived, the type $p(2,2)$ never is, the type $p(2,1,1)$ — one double root, two simple — is the substantial positive theory, and the type $p(1,1,1,1)$ is the subject of Conjecture 1 of [2], which asserts that no quartic with four distinct roots is rational-derived and which remains open. (For quintics with a triple root the corresponding nonexistence was settled by Flynn [8].) Theorem 7 of [2] presents the full classification, and subsequent constructions [3, 5] cite it as the classification of the quartic case.

1.1. What this paper does.

What this paper does. This paper reports a three-part study of that classification, conducted through a Lean 4 [10] formalization, and its results sort the quartic picture into three strata. *The repeated-root stratum is now classified unconditionally*: Part II states and proves Theorem 7′ in image form — the normalised $p(2,1,1)$ sector is exactly the image of the rational points of the curve `576i2` under an explicit map, with no subgroup or coset restriction — closes the statement-level links (gate form against natural form, normalised family against arbitrary split quartics) in the kernel, and assembles the unconditional classification of all repeated-root quartics (Theorem 5.3). *The four-distinct-root stratum is mapped to its boundary*: Part I locates, by a unit-by-unit transcription of the published proof, the exact step of [2] that is not established — a printed assumption equivalent to the conclusion it should deliver, followed by a universal claim the preceding analysis does not cover — and refutes the reconstructed implication with a single quartic read over two base fields; Part III then proves, in the kernel, that Conjecture 1 is equivalent to the nonexistence of nondegenerate rational points on an explicit splitting surface, and shows that all three routes the printed argument could take are governed by double covers that are nontrivial over the splitting extension. *Conjecture 1 itself remains open*: nothing in this paper resolves it, and §8 displays it, deliberately unproved, in the exact form it takes in the development. Along the way the audit corrects two Mordell–Weil ranks in the sources (§3) and yields the historical unification above.

Throughout, "the formalization" names the whole program — the verbatim transcription and audit of the published text, the computer-algebra gating of every statement-level definition, the statement-level design pressure, and the kernel verification — not the kernel alone. The arc of the paper is that this program *detects, sharpens, and certifies*: the audit detected a gap that twenty-six years of use had not surfaced; the discipline of writing formal statements sharpened loose ones (the coset restriction of the 1995 model, the excluded parameters $a \in \{0, 1\}$, the degenerate loci of the a-map, all now exactly delimited); and the kernel certified the repair, unconditionally, with the standard three-axiom footprint (§7.3). Attribution is kept at that resolution deliberately: it would be inaccurate to say simply that "Lean found the gap."

1.2. **What we learned.** The lesson we consider most transferable is that a limitation of the library produced a better theorem. The published route to Theorem 7 runs through the Mordell–Weil rank of the curve; mathlib [13], at the pinned commit, did not provide the Mordell–Weil rank machinery required by the published route, so the conditional first version of our development carried the rank as a disclosed axiom. The pressure to remove it forced a different question — what do the two gate witnesses alone determine? — and the answer (§4.3) is an explicit rational reconstruction of a point of the curve from the witnesses, discovered by Gröbner elimination and certified by the kernel, which proves the completeness direction with *no rank input at all.* The repaired theorem is not merely formalized; it is unconditional in a way Theorem 7's own statement was not (the coset-form classification of [1], in the repeated-root sector, was itself unconditional; the conditionality sat in Theorem 7's exhaustion claim), and the disclosed axiom, deliberately retained, marks in the kernel's own per-declaration report the point where the condition was removed (§7.3).

This experience composes three lessons already recorded in the Annals of Formalized Mathematics. Loeffler and Stoll describe how a formal statement of the Riemann hypothesis nearly slipped at its edge — a junk value that, had it been zero, would have made the displayed statement a false proposition superficially resembling RH — and warn that statement-level slips are the failure mode to fear [11]; the present paper is the same failure type located in the informal literature, where it appears to have gone unnoted through twenty-six years of citation, and the statement-fidelity gates of §5 are our systematic response. Best, Birkbeck, Brasca and their coauthors describe formalizability driving the choice of proof, back to older and more elementary arguments [4]; here the drive went further, to a rank-free proof absent from the cited record. Ludwig and Merten describe formalization changing the mathematics — generalizations forced by the formal setting [12]; here the change is the unconditional two-gate form of the statement itself. The type of the present paper — the audit, repair and certification of a published, peer-reviewed classification — combines the three in a single audit-and-repair arc: the slip at the statement's edge is its subject, the return to the sound 1995 kernel its method, and the sharpened, rank-free theorem its outcome.

1.3. **The artifact.** Every kernel-verified statement cited in this paper is hyperlinked, at theorem level, to a specific declaration at the frozen release v1.1.0 (commit `a5cf446`) of `https://github.com/hirokifukui/rd-quartics-classification`; load-bearing computer-algebra claims are indexed to their scripts and captured logs (§7.2); the dependency graph with per-node kernel status is the blueprint at `https://hirokifukui.github.io/rd-quartics-classification/blueprint/`, and the archived snapshot carries concept DOI `10.5281/zenodo.21766032`. The live development comprises 4,347 lines of Lean across twelve files in two components — `Gap/` (the audit and the boundary) and `Thm7Prime/` (the repair) — with 209 theorems, one disclosed axiom off the main line, and no `sorry`; toolchain, pins, continuous verification and quantitative details are §7. Part I is §§2–3; Part II is §§4–5; Part III is §6; §7 records the formalization and the division of labour, and §8 the open sector.

Part I. The audit

2. The gap

The object of this part is a single section of [2]: §2.2.4, "Vertical translation of a quartic," which supplies the last inferential step of that paper's Theorem 7. The audit is a survey of one section, not a verdict on the paper: the identities established there are exact and are re-proved in the kernel below, and typographical errors and inferential gaps are distinct phenomena, kept distinct throughout. For the audit the section was segmented into eleven inferential units T0–T10 covering it with no residue; each unit is transcribed as a Lean object in `Gap/BMThm7Transcript.lean`, over a general field (linearly ordered where an order is needed), so that a published K-derived quartic can act as a transparent negative control below. Line references $\ell.n$ are to the archived working layout of the text (hash in `Gap/PROVENANCE.md`); that layout derives from the authors' 2009 revision, whose §2.2.4 differs from the published version only typographically, and all quotations below are given in the published wording.

2.1. **The promise and the assumption, quoted.** The section opens with its promise (ℓ.529–534):

> "If we now allow our rational-derived quartics to undergo a vertical translate by a rational distance, so that the (highest) local minimum, $x_{\min}$ say, is moved up to become a double root, then the remaining two roots, r and s say, of the quartic could possibly lie in a quadratic extension of $\mathbb{Q}$. Certainly, we have no a priori reason to expect the extra roots to be rational (see Fig. 2). None-the-less, in this section we show that the latter is precisely the case."

This is the terminus of the transcript: the proposition the section undertakes to prove, transcribed as `T0_claim`. The argument then names its standing datum. Having introduced $f(x) = x(x-1)(x-a)(x-b)$ and a rational translate amount c, it writes (ℓ.553–558):

> "We assume that the resulting quartic, $F(x)$ say, given by
>
> $$F(x) = x(x-1)(x-a)(x-b) + c \tag{4}$$
>
> is rational-derived and has a double root. As before, we require $\Delta(F) = 0$…"

(The equation is displayed there with tag (4); nothing else is elided.) The verb is *assume*; the assumption is transcribed as `T2_assumption`. And, having excluded the degenerate cases and established the discriminant identities recalled below, the section concludes verbatim (ℓ.615–620):

> "Previously one may have thought that the class of rational derived quartics with four distinct roots split into two types: those obtainable by a rational vertical translate from a p(2,1,1) quartic and those not so obtainable. The result of all the previous work shows that the latter class is empty and so any rational-derived quartic with four distinct roots can be vertically translated into a rational-derived quartic with a double root."

2.2. **The formal consequence: a circle and a quantifier jump.** The content of the printed assumption is fixed by a one-line observation.

Lemma 2.1 (`KDerived_translate_iff`). *Let $x_0 \in K$ be a critical point of the quartic $f \in K[x]$, and put $g(x) = f(x) - f(x_0)$. Vertical translation leaves every positive-degree derivative unchanged. Consequently, if f is K-derived, then g is K-derived if and only if g itself splits over K (the pinned kernel statement; equivalently, since $(x-x_0)^2$ always splits, if and only if the residual quadratic q in $g(x) = (x-x_0)^2 q(x)$ splits over K).*

Under the standing hypothesis of the section, the printed assumption is therefore *equivalent* to the missing splitting assertion for the selected critical translate — and in particular it already implies the existential conclusion asserted at the end of the section. The pointwise equivalence is kernel-certified

(`T2_iff_conclusion`). The structure of the argument is thus: (1) translate at the selected critical point (the highest local minimum); (2) assume the translate is rational-derived — by Lemma 2.1, assume the required splitting for that translate; (3) analyse, under that assumption, the resulting repeated-root family; (4) conclude, in the sentence quoted above, that the class not covered by the assumption "is empty", without an argument. The gap is a circularity together with a quantifier jump: the exhaustion is asserted for all rational-derived quartics, while the preceding analysis covers exactly those satisfying the assumption. The kernel transcript records this structurally: no lemma of the transcribed chain takes the splitting of f'' as a hypothesis — the full derived condition enters only through the assumption, which is precisely the pointwise splitting assertion required for the selected translate — and the concluding universal statement is derivable from the chain precisely when that assumption is supplied (`T9_via_T2`). The implication the section actually asserts — from the hypotheses consumed by the transcribed chain to the splitting of the translate — is transcribed, *as a definition and not a theorem*, as `T9_bridge`; it is refuted in §2.3 below.

The origin of the slip is instructive. For *cubics* the analogous bridging step is a theorem: after forcing a double root, a single root remains, and its rationality is free from the rationality of the root sum — this is exactly the argument of §2.1 of [2], which is sound. For quartics two roots remain, the root sum fixes only their sum, and the published argument carries the cubic conclusion across by analogy.

The correspondence between the published passages and the transcribed chain is summarized in the following table; the axiom footprint of every node is the standard three (§7).

Published passage	Formal node	Hypotheses consumed	Output	Role
assumption preceding (4)	`T2_iff_conclusion`	quartic is K-derived; selected critical point x_0	printed assumption ⇔ splitting of the residual quadratic at x_0	missing bridge
discriminant identities	`deltaF_eq_sym_mul_deltaF_cubed`, `D6_eq_sq_of_vieta`, `D6_nonneg_of_vieta`	local p-data; critical-point identities (Vieta data)	the displayed identities; $D_6 \geq 0$	non-circular chain
concluding paragraph	`T9_via_T2`	the printed assumption	splitting at the selected translate	quantifier jump

2.3. **One quartic over two base fields.**

Theorem 2.2 (One quartic, two base fields)**.** *Let*

$$f(x) = x(x-136)(x-52)(x+312), \qquad F(X) = 448^{-4} f(448X - 312),$$

so that $F(X) = X(X-1)\left(X - \frac{39}{56}\right)\left(X - \frac{13}{16}\right)$*, and let* $K = \mathbb{Q}(\sqrt{3441})$*. Then:*

(a) *over $\mathbb{Q}$, f has four distinct rational roots, f' splits, f avoids every symmetric configuration excluded in* [2, §2.2.4]*, and no critical vertical translate of f splits over $\mathbb{Q}$; hence the algebraic hypotheses actually used by the non-circular portion of the argument do not imply its concluding claim (`not_T9_bridge_Q`);*

(b) *F is properly K-derived with four distinct roots, and no critical vertical translate of F splits over K; hence the natural K-analogue of the bridging claim is false (`not_T0_claim_K` at the critical point selected by the published argument);*

(c) *F is the affine normalisation of the K-derived quartic exhibited by Stroeker* [15, Ex. 5.2] *at $t = -7/17$.*

The identity $F(X) = 448^{-4} f(448X - 312)$ carries the roots $0, 136, 52, -312$ of f to $\frac{39}{56}, 1, \frac{13}{16}, 0$; the two parts of the theorem are two readings of this single affine class. (Different normalisations of the class appear in the artifact: dividing the roots of f by 136 gives parameters $(13/34, -39/17)$ with selected critical point $x_0 = 13/17$, the coordinates used in the kernel certificates; the normalisation $(39/56, 13/16)$ used in the text is the one matching Stroeker's example. Throughout, "highest local minimum" abbreviates the transcript's algebraic second-derivative critical-point predicate, stated over a linearly ordered field; over $\mathbb{R}$, under the nondegeneracy hypotheses in force, it coincides with the higher of the two local minima.)

Over $\mathbb{Q}$. The roots $0, 136, 52, -312$ of f are distinct rationals, and

$$f'(x) = 4x^3 + 372x^2 - 103168x + 2206464 = 4(x + 221)(x - 24)(x - 104)$$

splits over $\mathbb{Q}$. In the normalised coordinates $(a, b) = (39/56,\ 13/16)$,

$$a - b - 1 = -\tfrac{125}{112}, \qquad a - b + 1 = \tfrac{99}{112}, \qquad a + b - 1 = \tfrac{57}{112},$$

all nonzero, so f avoids every symmetric configuration excluded in the argument, and the discriminant identities of §2.2 hold for it as identities. Since $f'(x_0) = 0$, each $f(x) - f(x_0)$ has a double root at x_0; dividing by $(x - x_0)^2$ leaves a quadratic q_{x_0}:

$$\begin{array}{llll} x_0 = -221: & q(x) = x^2 - 318x + 40131, & \text{disc} = -59400 & = -66 \cdot 30^2; \\ x_0 = 24: & q(x) = x^2 + 172x - 43904, & \text{disc} = 205200 & = 57 \cdot 60^2; \\ x_0 = 104: & q(x) = x^2 + 332x + 6656, & \text{disc} = 83600 & = 209 \cdot 20^2. \end{array}$$

Neither 57 nor 209 is a square, and the first discriminant is negative; hence no critical translation of f has rational remaining roots. Each identity above is a polynomial identity over $\mathbb{Z}$ and can be checked directly; the full chain (the factorization of f', the three divisions, the three discriminants, and the non-squareness assertions) is certified by the kernel under the standard three axioms (`Gap/BMThm7Gap.lean`), unconditionally over $\mathbb{Q}$. We note that f itself is not rational-derived: $\operatorname{disc}(f'') = 40^2 \cdot 3441$. This is exactly what part (a) requires — the counterexample must live outside the circular assumption while satisfying all algebraic premises actually invoked after its removal — and it is what part (b) turns to advantage.

Over $K = \mathbb{Q}(\sqrt{3441})$. Over K the second derivative also splits:

$$F''(X) = \frac{5376X^2 - 6744X + 1859}{448}, \qquad \text{with roots } \frac{843 \pm 5\sqrt{3441}}{1344} \in K,$$

so F is K-derived, and properly so, since it is not rational-derived. Its critical points are given by

$$F'(X) = \frac{(4X-3)(14X-13)(64X-13)}{896}, \qquad X_0 \in \left\{\tfrac{13}{64},\ \tfrac{3}{4},\ \tfrac{13}{14}\right\},$$

and the residual discriminants of the three critical translates are

$$-66\left(\tfrac{15}{224}\right)^2, \qquad 57\left(\tfrac{15}{112}\right)^2, \qquad 209\left(\tfrac{5}{112}\right)^2$$

— the same square classes as over $\mathbb{Q}$, as they must be, this being the same quartic. Now K is real, so the first is not a square in K; and a rational number r is a square in K only if r or $3441r$ is a square in $\mathbb{Q}$, which fails for 57 and 209 (neither $57 \cdot 3441 = 196137$ nor $209 \cdot 3441 = 719169$ is a square). Hence *no* critical vertical translate of F splits over K, although F satisfies the full K-derived hypothesis. The statement at the critical point selected by the argument ($X_0 = 13/14$, the highest local minimum, with residual quadratic $X^2 - \frac{73}{112}X + \frac{13}{6272}$) is kernel-certified over an explicit subfield model of K (`not_T0_claim_K`); the failure at the remaining two critical points is by the direct computation above. Both refutations, over $\mathbb{Q}$ and over K, funnel through the same square class 209.

Two statements should be kept separate. Theorem 2.2 does not assert that the statement of [2, Thm. 7] over $\mathbb{Q}$ is false — a counterexample over $\mathbb{Q}$ would refute Conjecture 1 of the same paper, which is open (§8). What fails is the passage from the reduction to the stated exhaustion; whether that passage holds over $\mathbb{Q}$ under the full rational-derived hypothesis is precisely the unresolved classification question, entangled with Conjecture 1. Part (b) shows that in the quadratic-field setting to which the paper extends the theory, the analogous passage fails as such. As a statement about the literature: we found no independent proof or published repair in the sources and citation chains examined through July 2026; the exhaustion over $\mathbb{Q}$ must accordingly be regarded as unproved.[1]

2.4. **Provenance, the surviving parts, and the replacements.** The affine class of the counterexample is not new. In [15, Ex. 5.2] Stroeker exhibits proper K-derived quartics with parameter $U(t) = (2t^2+2t-1)/(5t+4)$; at $t = -7/17$ (listed there with $D = 3{\cdot}31{\cdot}37 = 3441$) one has $U = -13/17$ and roots $1, -51/13, -17/7, -3$, which the affine map $X = 13(1-u)/64$ carries to $0, 1, 39/56, 13/16$ — the quartic F of Theorem 2.2. What is new here is the observation that this known K-derived quartic, read over $\mathbb{Q}$, satisfies all algebraic premises actually invoked by the non-circular portion

[1] A revised version of the paper, dated June 21, 2009, is available from the first author's website (`https://www.ralphthetriangle.com/papers/rational-derived-polynomials`; accessed July 2026); its §2.2.4 is unchanged.

of [2, §2.2.4] after the printed assumption is removed, while falsifying the concluding implication of that portion; and that over K it refutes the natural K-analogue of the bridge outright. A witness to the gap in the proof of the family's central classification was thus already present in the family's own literature.

The audit is a survey, not a demolition, and it closes with an accounting of what survives and what was replaced. The translate setup is sound. The displayed discriminant identities are exact: the identity $\Delta(f') = -16\,D_6(a,b)$ at $\ell.606$ and the -2^{12} form of ΔF at $\ell.568$ hold as printed (here Δ denotes the resultant-normalised convention of that paper, to be distinguished from the ordinary polynomial discriminant, which satisfies $\operatorname{disc}(f') = +4\,D_6$ — kernel-proved at the level of the closed formulas, with the coefficient bridge; the resultant-convention identifications themselves are certified in both computer-algebra systems [C]). Within the kernel the closed forms are reconciled: `deltaF_eq_sym_mul_deltaFp_cubed` proves that the closed form of ΔF equals sym$\cdot$(closed form of $\Delta(f')$)^3 with prefactor exactly 1 — a tower nowhere printed with that prefactor. Of the two mutually inconsistent printed prefactors (-2^{12} against D_6^3 at $\ell.568$ and 2^8 against $\Delta(f')^3$ at $\ell.611$, which with $\Delta(f') = -16D_6$ contradict one another), the reconciliation adjudicates the 2^8 as a typographical error, immaterial to the argument. Three further local repairs are recorded, each replacing a printed argument by a stronger, kernel-checked one. First, the printed stationary-point analysis of the surface $z = D_6(a,b)$ inspects three stationary points and two sample values and cannot by itself establish global non-negativity on an unbounded domain; the conclusion $D_6 \geq 0$ is nonetheless exact and is re-proved in analysis-free form, valid over any linearly ordered field: on the Vieta locus D_6 is 64 times a square (`D6_eq_sq_of_vieta`, `D6_nonneg_of_vieta`). Second, the disposal of the three symmetric configurations, attributed in the paper to a closed-access source, is replaced by a self-contained proof: in each symmetric case K-derivedness forces a rational point on the conic $r^2 = 3(p^2+q^2)$, which has only the trivial solution by 3-adic descent (`symF_zero_not_KDerived`), eliminating the external citation from the load path. Third, a "Conversely" assertion in the same section does not follow from the preceding identities; it is not used in the sequel, and no repair is required.

Finally, the audit was conducted under a pre-registration discipline: the segmentation into units, the classification of each unit, and seven numbered predictions about where the argument would and would not formalise were fixed in a document whose hash was frozen before any Lean build was attempted, and the adjudication record postdates it in full. All seven predictions were confirmed; two were confirmed in a form stronger than predicted. Predictions and confirmations were both produced within this project, so the record is presented as procedural transparency, not as independent verification.

3. Two corrected ranks and the curve E_0

The audit also recomputed the numerical curve data of [2] and [15]. Two entries required correction; a single curve turned out to organize the family's whole record. The computations of this section are retained as scripts and logs in the archived development (`scripts/rank/`, a SageMath 10.8 audit; the rank of E itself is additionally certified in Magma, `scripts/magma/`). The rank statements of Theorem 3.1 are unconditional (2-descent; no analytic input is consumed by any statement of this paper).

Theorem 3.1 (Corrected ranks)**.** *Let* $E\colon z^2 = w(w-6)(w+18)$ *be the auxiliary curve of* [2]*.*

(a) $\operatorname{rk} E(\mathbb{Q}(\sqrt{33})) = 2$*, not* 1 *as recorded in* [2, Table 5]*.*
(b) *The elliptic curve underlying Example 4.1 of* [15] *(identical in the earlier report* [14]*) has Mordell–Weil rank* 1 *over* $\mathbb{Q}$*, not* 2 *as stated there.*

(a) Table 5, $m = 33$*.* With E_m the quadratic twist of E by m (for $m = 33$, minimal model $y^2 = x^3 - x^2 - 2097x + 28305$), and the twist identity $\operatorname{rk} E(\mathbb{Q}(\sqrt{m})) = \operatorname{rk} E(\mathbb{Q}) + \operatorname{rk} E_m(\mathbb{Q})$ used throughout [2, §4]: unconditional two-descent gives $\operatorname{rk} E(\mathbb{Q}) = 1$ and $\operatorname{rk} E_{33}(\mathbb{Q}) = 1$, the latter with explicit generator $(-35, 240)$ on the minimal model, saturation index 1 (the analytic rank agrees, as an auxiliary check); hence $\operatorname{rk} E(\mathbb{Q}(\sqrt{33})) = 2$. A direct computation over $\mathbb{Q}(\sqrt{33})$ independently returns rank 2, with the two independent generators $(-12, 36)$ and $(3-3s,\ 36-12s)$, $s = \sqrt{33}$, on E. The $m = 33$ entry is the sole discrepancy we assert. The row-by-row audit retained in the artifact is partial and not load-bearing: seven of the nine entries of Table 4 are confirmed there by unconditional two-descent and the remaining two agree with the analytic ranks reported in the source; of Table 5, the $m = 73$ row is confirmed unconditionally and the $m = 57$ row analytically only; the sweep, including its aborted run, is preserved as such in `scripts/rank/`.

(b) Example 4.1 of [15]*.* The curve of that example is the quartic model

$$Z^2 \;=\; 3\,(t^4 + 4t^3 + 4t^2 + 3),$$

with rational points (e.g. $(t, Z) = (0, \pm 3)$), whose Jacobian is $\mathbb{Q}$-isomorphic to E_0 below; by the rank of the curve we mean the Mordell–Weil rank of that Jacobian. Both versions state that this curve has rank 2. Unconditional two-descent gives rank bounds $(1, 1)$, hence the Mordell–Weil rank is 1; the point $(-8, 36)$ generates $E_0(\mathbb{Q})$ modulo its torsion (saturation index 1, checked unconditionally; log archived with the release accompanying this paper). The analytic rank is also 1, an independent consistency check. The example's substantive conclusion is unaffected: the correspondence asserted there is independent of the rank computation, and rank 1 still makes the Mordell–Weil group infinite, so the infinitude of examples is preserved. (The result first circulated as Econometric Institute Report EI 2002-30 (2002) and was then published in condensed form as [15]; Example 4.1, with the rank-2 statement, appears identically in both versions. The correction applies to both; we address it primarily to the published version.)

Theorem 3.2 (One curve through the family)**.** *Let* $E_0\colon y^2 = x^3 - 156x + 560$, *of conductor* 576*. Then:*

(a) E_0 *is* $\mathbb{Q}$*-isomorphic to the auxiliary curve* $E\colon z^2 = w(w-6)(w+18)$ *of* [2]*, so that Tables 4 and 5 of that paper are rank tables for the quadratic twists of* E_0*;*

(b) E_0 *is* $\mathbb{Q}$*-isomorphic to the Jacobian of the quartic model in Example 4.1 of* [15]*; in particular the corrected rank of Theorem 3.1(b) is the rank of* $E_0(\mathbb{Q})$ *itself, which is* 1*;*

(c) *let* $P = (-12, 36)$ *on* $E\colon z^2 = w(w-6)(w+18)$*, a point of infinite order generating* $E(\mathbb{Q})$ *modulo its torsion* $(\mathbb{Z}/2\mathbb{Z})^2$*, and let* $a(w,z)$ *be the map* (4)*. Then*

$$a(2P) = \frac{90}{77}, \qquad a(3P) = \frac{497610}{167167},$$
$$a(4P) = -\frac{128665027260}{69283212011}, \quad a(5P) = \frac{4173952380480465660}{7010366636418797651},$$

and these are, in order and up to the root-exchange symmetry $a \leftrightarrow 1/a$*, the normalised parameters of the quartics* $x^2(x-p)(x-q)$ *of the four pairs* (p,q) *printed in Guy's problem column* [9] *— the fourth pair after exchanging the two simple roots. The first pair* $(308, 360)$ *yields a translate of Klamkin's quartic* $(x-193)(x-141)(x+167)^2$ *(circa 1950; see* [6]*). (Guy separately credited M. J. Miller with a refutation of Carroll's uniqueness conjecture; Miller's example is not printed there, and we make no identification of it.)*

For (a) no computation is needed: the substitution $w = x - 4$ carries $z^2 = w(w-6)(w+18)$ to $y^2 = (x-4)(x-10)(x+14) = x^3 - 156x + 560$, an identity the reader can check directly; both curves have conductor 576 (E_0 is `576i2` in Cremona's tables — the very curve of Theorem 7′). For (b), the quartic $Z^2 = g(t)$ with $g(t) = 3t^4 + 12t^3 + 12t^2 + 9$ has classical invariants $I = 12ae - 3bd + c^2 = 468$ and $J = 72ace + 9bcd - 27ad^2 - 27b^2e - 2c^3 = -15120$ (writing $g = at^4 + bt^3 + ct^2 + dt + e$; for this convention and the Jacobian $y^2 = x^3 - 27Ix - 27J$ see, e.g., [7]), hence Jacobian $y^2 = x^3 - 12636x + 408240$; setting $x = 9X$, $y = 27Y$ and dividing by 3^6 gives $Y^2 = X^3 - 156X + 560$, i.e. E_0. Both steps can be checked by hand. For (c): translating Klamkin's quartic $p(x) = (x-193)(x-141)(x+167)^2$ — whose derivative has the integer roots $\{169, -2, -167\}$ and whose second derivative has roots ± 97 — by 167 gives $x^2(x-308)(x-360)$, the quartic of Guy's first printed pair; normalizing to the form $x^2(x-1)(x-a)$ gives $a = 360/308 = 90/77$. Guy's second printed pair $(668668, 1990440)$ equals $4 \cdot (167167, 497610)$, and its quartic — first-derivative roots $\{0, 423696, 1570635\}$, second-derivative roots $\{195624, 1133930\}$, all integers — normalizes to $a = 497610/167167$

(lowest terms). With $P = (-12, 36)$ (equivalently $(-8, 36)$ on E_0 under $w = x - 4$), direct computation of the multiples gives the four parameters displayed, with both gates of (2) satisfied at each of $2P$, $3P$, $4P$, $5P$ (script and log archived with the release accompanying this paper). Guy's third printed pair $(-277132848044,\ 514660109040)$ normalizes to $a(4P)$ exactly; his fourth,

$$(16695809521921862640,\ 28041466545675190604),$$

normalizes to $1/a(5P)$, i.e. to $a(5P)$ after exchanging the two simple roots. The four printed quartics are thus, up to that symmetry, the images of the four consecutive multiples $2P$ through $5P$.

The family's record from 1950 to 2019 — Klamkin's isolated example, Carroll's uniqueness conjecture [6], the pairs printed in Guy's column [9] (with a refutation credited separately to M. J. Miller, whose example we do not identify), the table of [1] — which already lists these four quartics as the consecutive multiples $n\,(75, 405)$, $n = 1, \ldots, 4$, on E_A, identifying the first with Carroll's example, the organizing observation in print — the rank tables of [2], and the infinite families of [3, 5] — may be viewed retrospectively as the discovery, one multiple at a time, of the images of a single point on a single curve. The steps added here are the identification with Guy's printed pairs, the transport to the single curve E_0, and the identification of Stroeker's Jacobian (Theorem 3.1).

Part II. The repair

4. Theorem 7′: the unconditional repair

Part I located the unproved step; this part states the classification that the evidence supports and proves it unconditionally. The 1995 kernel of the result — the description, due to Buchholz and Kelly [1], of the family $x^2(x-1)(x-a)$ — is sound, and the gap lies precisely in the 2000 increment; Theorem 7′ below is its constructive counterpart. Two features of the proof are worth announcing. First, *no rank computation enters*: the published treatment proceeds via the Mordell–Weil rank of the curve, and we show that the rank is needed only to *enumerate* its rational points, not to establish the classification itself. Both directions reduce to explicit polynomial certificates together with elementary field and order arguments. Second, the entire theorem is formalized: the main statement `thm7prime` compiles against mathlib with the standard three-axiom footprint (§7). The main text records each identity and the logical role of each certificate; the archived development carries the complete coefficient lists.

4.1. **The normalised family and the statement.** A quartic of type $p(2, 1, 1)$ with double root c and simple roots $r_1 \neq r_2$ is carried by the affine substitution $x = c + (r_1 - c)u$ to a nonzero multiple of the *normalised* *family*

$$q_a(x) \;=\; x^2(x-1)(x-a), \qquad a = \tfrac{r_2-c}{r_1-c}, \tag{1}$$

with $a \notin \{0,1\}$; rational-derivedness is preserved by such substitutions. (This normalisation identity is itself one of the formalized lemmas, `caseA_normalized`; the full bridge from an arbitrary split $(2,1,1)$ quartic is the affine classification of §5.) Since $q_a = x^4 - (1{+}a)x^3 + ax^2$, we have

$$q_a' = x\,(4x^2 - 3(1{+}a)x + 2a), \qquad q_a'' = 12x^2 - 6(1{+}a)x + 2a,$$

and q_a''' always splits. Hence q_a is rational-derived if and only if the two discriminant *gates*

$$G_1(a):\ 9a^2 - 14a + 9 = r^2, \qquad G_2(a):\ 9a^2 - 6a + 9 = s^2 \tag{2}$$

admit rational solutions r, s (`Gate1`, `Gate2`). Write $\mathrm{RD}(a)$ for the conjunction "$a \neq 0$, $a \neq 1$, $G_1(a)$, $G_2(a)$" (`RD211`); the equivalence of this gate form with the natural derivative-splitting form is itself a kernel theorem (§5).

On the curve

$$E:\quad z^2 \;=\; w^3 + 12w^2 - 108w \;=\; w(w-6)(w+18) \qquad \text{(Cremona } \mathtt{576i2}\text{)}, \tag{3}$$

the identity $(z-w-18)(8w+z) = (7w-18)z + (w^3+4w^2-252w)$ holds (`D_reduce`), and we write $D(w,z)$ for this common value of the denominator of the Buchholz–MacDougall map

$$a(w,z) \;=\; \frac{9\,(2w+z-12)(w+2)}{(z-w-18)(8w+z)} \tag{4}$$

(`aMap`). Here and below, $(w,z) \in E(\mathbb{Q})$ means an affine rational point of (3); the point at infinity, at which (4) is undefined, plays no role.

Theorem 4.1 (Theorem $7'$, image form (the normalised $p(2,1,1)$ sector); `thm7prime`)**.** *Let $a \in \mathbb{Q}$.*

- (i) (Backward.) *If $(w,z) \in E(\mathbb{Q})$ with $D(w,z) \neq 0$ and $a := a(w,z) \notin \{0,1\}$, then* $\mathrm{RD}(a)$.
- (ii) (Forward / completeness.) *If* $\mathrm{RD}(a)$*, then there exists $(w,z) \in E(\mathbb{Q})$ with $D(w,z) \neq 0$ and $a(w,z) = a$.*

Interchanging the two simple roots of (1) replaces a by $1/a$, so q_a and $q_{1/a}$ represent the same affine-equivalence class; Theorem 4.1 characterizes the complete set of normalised parameters, not a choice of representative for each class. The theorem is stated on $E =$ `576i2` with *no coset restriction*; the relation to the coset-restricted 1995 statement is §4.4. The Weierstrass model is tied to mathlib's own vocabulary inside the development (`OnE_iff_equation`, `E576i2_elliptic`).

4.2. **The backward direction: gate identities.** The backward direction is a pair of square identities in the function field of E. Set $M(w) = w(w+18)(w^2 - 63w + 486)$; note $w^2 - 63w + 486 = (w-9)(w-54)$. On E, away from the loci $M(w) = 0$ and $D(w,z) = 0$, one has the *exact* identities

$$9a^2 - 14a + 9 = \left(\frac{3(w-18)(w+6)w(w+18) + (-21w^2 + 108w - 2916)\,z}{M(w)}\right)^2, \tag{5}$$

$$9a^2 - 6a + 9 = \left(\frac{3(w-18)(w+6)w(w+18) + (-9w^2 - 648w + 2916)\,z}{M(w)}\right)^2, \tag{6}$$

where $a = a(w,z)$. Cleared of denominators these are polynomial identities modulo the curve equation, verified in Lean by `linear_combination` with explicit cofactor certificates (`gate1_cleared`, `gate2_cleared`); the certificates were computed exactly in Sage and are reproduced verbatim in the formal development, so their correctness is checked by the kernel rather than trusted.

The excluded locus is finite and harmless: on $E(\mathbb{Q})$ the conditions $M(w) = 0 \neq D(w,z)$ hold only at $(54, 432)$ and $(9, -27)$, both of which map to $a = 77/90$, where the gates hold with the explicit witnesses $r = 19/10$ and $s = 97/30$ (`exceptional_points`). This proves Theorem 4.1(i) (`thm7_backward`).

4.3. **The forward direction: an explicit reconstruction from the gate witnesses.** Let a satisfy $\mathrm{RD}(a)$, with gate witnesses r, s as in (2); note the sign freedom $r \mapsto -r$, $s \mapsto -s$. Gröbner elimination shows that a point of E above a can be reconstructed rationally from the triple (a, r, s). Concretely: in $\mathbb{Q}[z, w, a, r, s]$ with lexicographic order, let I be generated by the curve equation, the graph relation $a\,D(w,z) - 9(2w + z - 12)(w + 2)$, and the relations $r\,M(w) - N_1(w,z)$, $s\,M(w) - N_2(w,z)$ tying the witnesses to the square roots of (5)–(6), with N_1, N_2 the numerators displayed there. After saturating I by $D(w,z)\,M(w)$, its Gröbner basis contains the linear relation $(3r - 7s + 12)\,w - 18\,(15a - 6r - s - 13)$. The saturated Gröbner-basis computation was performed in Magma; Sage independently verified the resulting reconstruction formulas and the polynomial identities used below. The Gröbner computation is used only for *discovery* — correctness of the reconstruction is established by the kernel-checked identities. Set

$$W = 18\,(15a - 6r - s - 13), \qquad T = 3r - 7s + 12, \qquad w = \frac{W}{T} \tag{7}$$

(`wNp`, `wDp`), and determine z linearly from $a \cdot D(w,z) = 9(2w + z - 12)(w + 2)$: explicitly $z = z_N / z_D$ with

$$\begin{aligned} z_N &= 9\,(2W - 12T)(W + 2T)\,T - a\left(W^3 + 4W^2T - 252\,WT^2\right), \\ z_D &= 18\,K\,T^2, \end{aligned} \tag{8}$$

where

$$K(a,r,s) \;=\; 105a^2 - 45ar - 238a + 51r + 16s + 105 \tag{9}$$

(`zNp`, `zDp`, `K1p`). Two polynomial consequences are then kernel-checked modulo the two gate relations, each by a single `linear_combination` step with a Sage-derived cofactor certificate: `curve_cleared` proves that the constructed point lies on E, and `aden_cleared` proves the pullback identity

$$D(w,z)\cdot T^3 z_D \;=\; -62208\,P(a,r,s) \tag{10}$$

for the explicit sextic P of Appendix A (`adP6`), so that $P(a,r,s)\neq 0$ together with $T\neq 0$ and $z_D\neq 0$ forces $D(w,z)\neq 0$. Independently, the definition of z gives the cleared graph identity $a\,D(w,z)=9(2w+z-12)(w+2)$; once the sign choice and (10) establish $D(w,z)\neq 0$, this identity yields $a(w,z)=a$.

Three denominators must be controlled, and this is where the sign freedom is spent.

(a) *The w-denominator.* Write $Q(a)=225a^2-210a+238=(15a-7)^2+189>0$. An explicit elimination certificate gives $a^2Q(a)\in\langle\, T,\ r^2-(9a^2-14a+9),\ s^2-(9a^2-6a+9)\,\rangle$, so for $a\neq 0$ the gate relations force $T\neq 0$, for *every* choice of signs (`wD_ne`).

(b) *The factor K.* A second elimination certificate gives $(3a-5)(5a-3)\,a^2Q(a)\in\langle\, K,\ r^2-(9a^2-14a+9),\ s^2-(9a^2-6a+9)\,\rangle$. At $a=5/3$ the first gate forces $(3r)^2=96$, and at $a=3/5$ it forces $(5r)^2=96$; since 96 is not a rational square, these branches are vacuous, and again $K\neq 0$ for every choice of signs once $a\neq 0$ (`K1_ne`, `sq_ne_96`).

(c) *The a-map denominator at the constructed point.* By the pullback identity (10), this reduces to $P(a,r,s)\neq 0$. The ideal generated by the two gate relations together with the four sign conjugates $P(a,\pm r,\pm s)$ is the *unit ideal*: a Nullstellensatz certificate (six cofactors, reproduced in the formal development) exhibits 1 as a combination, so at least one sign choice keeps $P\neq 0$ (`adP6_cover`).

Choosing signs by (c) and applying (a), (b) yields a point of $E(\mathbb{Q})$ off the degenerate locus with $a(w,z)=a$ (`forward_core`, `thm7_forward`), proving Theorem 4.1(ii). For example $a=90/77$, with $r=171/77$, $s=291/77$, yields $(w,z)=(726/25,\ 22176/125)$.

Remark 4.2 (No rank input)**.** *The published treatment of Theorem 7 proceeds through the Mordell–Weil rank of E (equal to 1; re-verified unconditionally by 2-descent in two independent systems). Theorem $7'$ consumes no rank fact in either direction: rank enters only if one wishes to enumerate $E(\mathbb{Q})$ and hence list the rational-derived parameters, as in* [1]*. In the formal development the rank-1 statement is present as a single disclosed axiom, instance-localized to the curve (`rank_E576i2`), retained for that enumerative purpose and consumed only by auxiliary declarations from an earlier conditional architecture; the kernel's per-declaration axiom report records that the main theorem does not depend on it (§7).*

4.4. **Relation to the 1995 classification.** Theorem $7'$ is not a transport of the theorem of [1] across the isogeny of Remark 4.3, for two reasons worth making explicit. First, the 1995 classification runs through the subgroup structure of the model curve `576i3`: the set of points corresponding to rational-derived quartics is the index-two subgroup generated by $(75, 405)$ and $(-6, 0)$ (their Theorem 1), the parameter map $(X, Y) \mapsto (5X+Y+30)/(9(X+2))$ is injective on the infinite cyclic subgroup generated by $(75, 405)$ (their Lemma 4), and the passage between the two cosets is the substitution symmetry $a \mapsto 1/a$ of the family (their Lemma 3). Neither subgroup carries an intrinsic description in the statement; both are identified through the 2-descent and generator computation of that paper, so the classification, and any statement transported from it, consumes the Mordell–Weil enumeration of the curve. Theorem 7′ consumes no such input: on `576i2` with the map (4), no subgroup or coset restriction is needed, and the degenerate locus is finite and explicit. Second, the completeness direction of [1] passes through the quartic curve $Y^2 = (9X^2 - 14X + 9)(9X^2 - 6X + 9)$, on which only the *product* of the two gates is a square; deciding which of its points satisfy each gate individually is again the subgroup determination. The reconstruction (7) works directly from the pair of gate witnesses and is, to our knowledge, new. What Theorem 7′ shares with [1] is the normalised family and the two gates; what it removes is the dependence of the classification on the arithmetic of the curve.

The closest precursor of the rank-free route, however, is a sketch in [2] itself. Working over quadratic fields, §3 of that paper solves one gate by the chord method, substitutes the resulting parametrization into the other, and passes through Mordell's transformation to the same curve E and to precisely the map (4), asserting that the map provides "the correspondence between all k-rational points … and all k-derived quartics" before rank determination enters — for enumeration. Read literally, the asserted correspondence does not hold: the map is undefined where aden vanishes, its values 0 and 1 correspond to degenerate quartics (the $\mathbb{Q}$-fibres are computed in §5.3), and neither direction of the correspondence is argued there. Theorem 7′ can be read as the proved form of that sentence: both directions are established, the exceptional locus is exactly characterized, and the certificates are closed-form, hence kernel-checkable. The comparison also sharpens §3: that the rank should be needed only to *enumerate* is implicit in the presentation of [2] itself; what was missing was the proof.

Remark 4.3 (The two curves)**.** *The Buchholz–Kelly parametrization lives on* $E_A =$ *`576i3`*: $Y^2 = X(X-48)(X+6)$ *and is stated there with a two-coset restriction; the map* (4) *lives on* $E =$ *`576i2`* *and, by Theorem* 7′*, needs no restriction. The two are reconciled by the explicit* 2*-isogeny* $E \to E_A$*,*

$$(w, z) \;\longmapsto\; \Big(\frac{(w+6)^2}{w-6},\; \frac{z\,(w+6)(w-18)}{(w-6)^2}\Big),$$

with kernel $\{O, (6, 0)\}$*, verified independently in Magma and Sage. The coset restriction appearing in* [1] *is a phenomenon of their model and their parameter map, not of* (3)–(4)*.*

Remark 4.4 (Errata in the sources)**.** *For the convenience of readers of* [1] *we record, verified by direct recomputation in the computer-algebra layer* *(script and log archived with the release accompanying this paper): rows* $n = 2, 3$ *of the table of* [1] *(p. 131) are internally inconsistent — the printed points and the printed a-values correspond to opposite multiples* $\pm nQ$ *of the printed generator* $Q = (75, 405)$*: the printed points equal* $-2Q$ *and* $-3Q$*, while the printed a-values and root pairs correspond to* $+2Q$ *and* $+3Q$*; no identification* $(X, Y) \equiv (X, -Y)$ *that would reconcile them is stated in that paper. The misprinted constant of* [2] *is §2.4; its two corrected Mordell–Weil ranks are §3. None of this affects the mathematics above.*

5. Statement fidelity and the affine classification

The kernel guarantees that a formal proposition follows from the axioms; whether that proposition models the mathematical problem is outside the kernel. This correspondence — what the AFM's editorial guidance calls *proof auditing* — was treated in this project as a gate: every statement-level definition was cross-checked in Sage and Magma against independent reimplementations *before* any proof was attempted, and the two links most exposed to a statement-level slip were subsequently closed in the kernel as well. This section records them, together with the elementary sectors and the sanity controls.

5.1. **Gate form and natural form.** The gate form $\mathrm{RD}(a)$ of §4.1 was chosen at design time as the division-free, verification-friendly restatement of rational-derivedness. Its correspondence with the natural form — the polynomial $Q_a = X^2(X-1)(X-\mathrm{C}\,a)$ and its first three `Polynomial.derivatives` split over $\mathbb{Q}$ (`NaturalRD`) — is a kernel theorem:

Theorem 5.1 (Gate form $\iff$ natural form; `rd211_iff_natural`)**.** $\mathrm{RD211}(a) \iff a \neq 0 \land a \neq 1 \land \mathrm{NaturalRD}(a)$.

Forward, the kernel-computed derivatives split explicitly under the gates; conversely, a quadratic over $\mathbb{Q}$ with nonzero leading coefficient that splits has square discriminant, applied to both derivatives. The CAS statement-fidelity gate remains in the development as the independent design-time cross-check (`scripts/sage/wp1_fidelity.sage`).

5.2. **The affine classification.** The reviewer-facing form of the repaired result is the bridge from an arbitrary split $(2, 1, 1)$ quartic to the normalised family, packaged as one kernel theorem. With $f \sim g$ iff $g = \nu\, f(\lambda X + \mu)$ for $\lambda, \nu \neq 0$ — an equivalence relation, kernel-checked (`AffineEquiv`) — and f a split $(2, 1, 1)$ quartic (`IsP211`):

Theorem 5.2 (Affine classification; `classification`, `classification_by_curve`)**.** *For a split* $(2, 1, 1)$ *quartic* f*:* f *is rational-derived iff* $f \sim Q_a$ *for some* a *with* $\mathrm{RD211}(a)$*; equivalently, iff* $f \sim Q_{a(w,z)}$ *for an affine rational point* (w, z) *of the Weierstrass model of* ***576i2*** *with* $D(w, z) \neq 0$ *and* $a(w, z) \notin \{0, 1\}$*.*

Rational-derivedness is invariant under $\sim$ (chain rule and transport of splitting along linear substitutions; $\sim$ is the equivalence relation induced by the action of the group $\langle X\rangle$ of [2] — the conservative vertical translations extending it to $\langle X^*\rangle$ are exactly the subject of §2); the normalisation $a = (r_2 - c)/(r_1 - c)$ is the kernel form of `caseA_normalized`; the gates then come from Theorem 5.1, and the curve form from Theorem 4.1.

5.3. **The excluded parameters and the elementary sectors.** The exclusion $a \neq 0$ is genuinely needed and completely understood: the fibre $a^{-1}(0) \cap E(\mathbb{Q})$ consists of the three points $(6, 0)$, $(-12, 36)$, $(-2, -16)$, corresponding to the degeneration of the family to the (always rational-derived) type $p(3, 1)$. The exclusion $a \neq 1$ is vacuous on rational points: an exact divisor computation shows the fibre $a^{-1}(1)$ has *no* rational points (the two quadratic factors of its eliminant $(w + 2)^2(w^2 - 36w - 108)(w^2 - 108)$ have no rational roots, and at $w = -2$ the graph relation forces $z = 16$, where $D(-2, 16) = 0$, outside the domain of the map); independently, $a = 1$ fails the second gate since 12 is not a square. The elementary sectors are dispatched directly: types $p(4)$ and $p(3, 1)$ are always rational-derived (up to the substitutions of §4.1 these are x^4 and $x^3(x - b)$, whose derivatives have visibly rational roots), and type $p(2, 2)$ never is (for $x^2(x - t)^2$ one finds $\mathrm{disc}(q'') = 48t^2$, and 48 is not a square). The pieces assemble into the full statement:

Theorem 5.3 (Theorem 7′, assembled form: the repeated-root classification)**.** *A quartic over $\mathbb{Q}$ with a repeated root is rational-derived exactly in the following cases: types $p(4)$ and $p(3, 1)$, always; type $p(2, 2)$, never; and, for a split $(2, 1, 1)$ quartic, if and only if it is $\sim$-equivalent to $Q_{a(w,z)}$ for an affine rational point (w, z) of* `576i2` *with $D(w, z) \neq 0$ and $a(w, z) \notin \{0, 1\}$.*

Each clause decomposes into kernel theorems: the $p(2, 1, 1)$ clause is Theorem 5.2, through Theorem 5.1 and the image form Theorem 4.1; the elementary sectors are the computations above; the case split by multiplicity type, and the observation that a quartic with an irrational root is not rational-derived (so the split hypothesis is harmless), are elementary [M]. The remaining sector — four distinct roots — is Part III.

5.4. **Counterexamples and sanity controls.** Following the practice of exhibiting the counterexamples that delimit a formalized statement, the development carries a sanity layer, kernel-checked independently of the main proofs: a positive control (the parameter $a = 77/90$ passes both gates, `gates_at_77_90`) and a near-miss that doubles as the negative control (Stroeker's properly K-derived quartic of Theorem 2.2: its f' splits over $\mathbb{Q}$ while f'' does not, `fp_splits`, `fpp_not_split_over_Q`, so the quartic passes the first gate over $\mathbb{Q}$, fails the second, and the pair of gates — not either alone — carries the classification). The same quartic serves as the transparent negative control of the transcript (§2), read over two base fields; the sanity layer and the audit share their witnesses by design, so that a statement-level slip in either would have been caught by the other.

Part III. The boundary

6. The terrain and the normalisation bridge

With Theorem 7′ the repeated-root sector is classified unconditionally; what remains open of the original Theorem 7 is exactly its $p(1,1,1,1)$ clause, entangled with Conjecture 1 of [2]. This part maps the terrain surrounding that boundary. Throughout it, each load-bearing claim carries a label: [K] = kernel-verified in Lean; [C] = computer-algebra computation (two-system where so stated; scripts and coverage in `scripts/CERTIFICATES.md`); [M] = ordinary mathematical deduction, written but not kernel-checked; [A] = research assessment. Narrative sentences without a label carry no evidential weight.

6.1. **The chart, the splitting surface, and the three covers.** Fixed notation for the whole part. B denotes the rational (a,b)-chart of the $\sigma_3 = 0$ root space of §2.2.3 of [2]: the quartic has roots $\{1, a, b, c\}$ with $c = -ab/\mathrm{den}$, $\mathrm{den} = ab + a + b$; its function field is $K(B) = \mathbb{Q}(a, b)$. The chart quantities (`den`, `s1n`, `Rq`, `Sq`, `Q4`) are $s_{1n} = \sigma_1\,\mathrm{den}$, $s_{2n} = \sigma_2\,\mathrm{den}$, the cleared splitting conditions $R_q = \mathrm{den}^2(9\sigma_1^2 - 32\sigma_2)$ (for f') and $S_q = \mathrm{den}^2(9\sigma_1^2 - 24\sigma_2)$ (for f''), and the funnel $Q_4 = (3R_q - S_q)/18 = \mathrm{den}^2(\sigma_1^2 - 4\sigma_2)$. Three kernel identities organize everything below:

$$18\,Q_4 = 3R_q - S_q \qquad \text{(I1, \texttt{I1\_cleared})}, \tag{11}$$

$$s_{1n}^2 - R_q = -8\,Q_4 \qquad \text{(I2, \texttt{I2})}, \tag{12}$$

$$Q_4 = F_1\,F_2\,F_3 \qquad \text{(\texttt{Q4\_factor})}, \tag{13}$$

with $F_1 = ab + a + b - 1$, $F_2 = ab + a - b^2 + b$, $F_3 = a^2 - ab - a - b$. The kernel statements are the identities only; that Q_4 is squarefree and that each $F_i = 0$ is an irreducible plane curve of genus 0 are two-system computer-algebra facts [C], not part of the kernel statement. Three further kernel factorisations (`pairing1`, `pairing2`, `pairing3`) identify, away from the vanishing of explicit cofactors, the locus $Q_4 = 0$ — where Theorem 7's conclusion is algebraically forced — with the locus on which the quartic $\{1, a, b, c\}$ is symmetric under one of the three root pairings: Buchholz–MacDougall's own excluded case.

S denotes the *splitting cover* of B: the biquadratic cover obtained by adjoining z with $z^2 = R_q$ and w with $w^2 = S_q$; its function field is, by definition, $K(S) = \mathbb{Q}(a, b)(z, w)$. Concretely, let $U_B \subset B$ be the complement of the vanishing locus of

$$\mathrm{den} \cdot ab\,(a-1)(b-1)(a-b)\,(ab + \mathrm{den})\,(a + \mathrm{den})\,(b + \mathrm{den})$$

— exactly the base-variable inequalities of a nondegenerate terrain point (Proposition 6.1: the factors beyond den express, after clearing den, that the four roots $\{1, a, b, c\}$, $c = -ab/\mathrm{den}$, are pairwise distinct and nonzero).

The splitting surface over the chart is the affine model

$$S_U := \operatorname{Spec}\big(\mathcal{O}(U_B)[z,w]/(z^2 - R_q,\ w^2 - S_q)\big),$$

and we write $U := S_U$ for this open model. Every element of $K(S)$ is a $K(B)$-combination of the spanning set $\{1, z, w, zw\}$ [M]; it is a basis once Proposition 6.3 is in place. Kernel statements about $K(S)$ below are proved in span form — quantifying over *any* field equipped with an embedding of $K(B)$ and square roots of R_q, S_q — and reach all of $K(S)$ through that spanning set [M].

For a critical point x_0 of the chart quartic q, *the disjunct at* x_0 is the statement that the vertical translate $q - q(x_0)$ — the transcript's `bmTranslate`, the very operation the printed proof performs — splits over $\mathbb{Q}$. The critical points are 0 and $(3\sigma_1 \pm \rho)/8$ with $\rho = z/\mathrm{den}$; the corresponding disjuncts are D_0, D_+, D_- (`D0`, `Dplus`, `Dminus`). Exchanging the sign of the square root ρ exchanges D_+ and D_-; no conditions beyond den $\neq 0$ and the splitting witness $z^2 = R_q$ are folded into the predicates. Each disjunct is governed by a double cover of U:

disjunct	cover	radicand	exclusion
$x_0 = 0$	$\widetilde{C}_0 \colon v^2 = Q_4$	Q_4 (by I1)	[K] Thm 6.4
$x_0 = \frac{3\sigma_1+\rho}{8}$	$C_+ \colon v^2 = \gamma_+$	$\gamma_+ = s_{1n}(s_{1n} - z)$	[K] Thm 6.5
$x_0 = \frac{3\sigma_1-\rho}{8}$	$C_- \colon v^2 = \gamma_-$	$\gamma_- = s_{1n}(s_{1n} + z)$	[K] Thm 6.5

The two $C_\pm$ radicands are *conjugate*; neither is claimed to have square class $-2Q_4$ itself. Their product — equivalently their layer norm — has square class $-2Q_4$: $\gamma_+\gamma_- = s_{1n}^2(s_{1n}^2 - z^2) = -8s_{1n}^2 Q_4$ by (12), and it is this *norm class* that the kernel obstruction consumes. Three covers, one sextic, two obstruction routes. Each cover's branch locus is supported on the zero locus of its radicand on U, with ramification exactly at the codimension-one points where the radicand has odd valuation.

6.2. **The normalisation bridge.** In normalised form, Conjecture 1 asserts that no quartic $x(x-1)(x-a)(x-b)$ with four distinct roots is $\mathbb{Q}$-derived (`Conjecture1_normal`; stated as a definition, without proof — the open statement of §8). The chart of §6.1 is where the printed argument of §2.2.3–2.2.4 of [2] works; the following kernel theorem identifies the two pictures exactly.

Proposition 6.1 (Normalization bridge; `conjecture1_iff_terrain`)**.** *There exists a $\mathbb{Q}$-derived quartic with four distinct rational roots (the $\{0,1,a,b\}$ normal form of Conjecture 1) if and only if there exists a nondegenerate rational point of the splitting terrain: $(a,b,z,w) \in \mathbb{Q}^4$ with $z^2 = R_q(a,b)$, $w^2 = S_q(a,b)$, $\mathrm{den}(a,b) \neq 0$, and the four roots $\{1,a,b,c\}$, $c = -ab/\mathrm{den}$, pairwise distinct and nonzero. Equivalently: the normalised Conjecture 1 holds iff $S(\mathbb{Q})$ has no nondegenerate point. (The reduction of the unnormalised conjecture to the $\{0,1,a,b\}$ normal form is the affine action — ordinary mathematics [M]; only the normal form enters the kernel.)*

Both directions are kernel theorems, with every normalisation condition tracked (`kderived_of_terrain`, `terrain_of_kderived`). In outline: a terrain point yields the monic quartic $f = (x-1)(x-a)(x-b)(x-c)$ with $e_3 = 0$, so $f'(0) = 0$ and $f' = x(4x^2 - 3e_1x + 2e_2)$; the witnesses z, w split the quadratic factors of f' and f'' via $9e_1^2 - 32e_2 = (z/\mathrm{den})^2$ and $9e_1^2 - 24e_2 = (w/\mathrm{den})^2$, and an affine change carries f to the $\{0, 1, a', b'\}$ normal form. Conversely, a $\mathbb{Q}$-derived quartic in normal form has a rational critical point x_0, which is not a root (the roots are distinct, so f is squarefree and $f'(r) \neq 0$ at each root r); translating x_0 to the origin and scaling the nonzero root $-x_0$ to 1 produces roots $\{1, A, B, C\}$ with $e_3 = 0$, forces $\mathrm{den}(A, B) \neq 0$ (else $AB = 0$, contradicting nonzero roots), and the rational root differences of the split quadratic factors supply the witnesses $z := 4\,\mathrm{den}\,(r_+ - r_-)$, $w := 6\,\mathrm{den}\,(u_+ - u_-)$.

6.3. **The fibre correspondence and the radicand derivation.** The disjuncts and the covers are tied together fibrewise, in the kernel.

Proposition 6.2 (Fibre correspondence; `fibre_correspondence`)**.** *Over* $\mathrm{den} \neq 0$ *with* $z^2 = R_q$*:* D_0 *holds iff the fibre of the cleared model* $\widetilde{C}_0 \colon v^2 = Q_4$ *has a rational point, and* $D_\pm$ *holds iff the fibre of* $C_\pm \colon v^2 = \gamma_\pm$ *has a rational point. (The kernel hypotheses are exactly these two —* $\mathrm{den} \neq 0$ *and* $z^2 = R_q$*; no further nondegeneracy is needed, so the formal statement is slightly stronger than the chart discussion requires.)*

For D_0 (`D0_iff_disc`): since $\sigma_3 = 0$, the translate at 0 factors as $q - q(0) = X^2(X^2 - \sigma_1 X + \sigma_2)$, so D_0 holds iff $\sigma_1^2 - 4\sigma_2$ is a rational square — the fibre of the *chart* cover $C_0 \colon t^2 = \sigma_1^2 - 4\sigma_2$. The chart cover and the cleared model are then identified by `cleared_model_iff`: over $\mathrm{den} \neq 0$ the substitution $v = \mathrm{den}\, t$ is a bijection on rational solutions (multiplication by the nonzero square den^2; the identity (11) is consumed definitionally), so clearing the denominator introduces no spurious solutions and loses none. For $D_\pm$ (`Dplus_iff`, `Dminus_iff`) the radicand is *derived*, not posited: at $x_0 = (3s_{1n} \pm z)/(8\,\mathrm{den})$ the translate factors as $(X - x_0)^2(X^2 + BX + C)$ with $B = 2x_0 - \sigma_1$, $C = \sigma_2 + 3x_0^2 - 2\sigma_1 x_0$, the linear-coefficient match being exactly $q'(x_0) = 0$ (available from $z^2 = R_q$); the discriminant identity $B^2 - 4C = \sigma_1(\sigma_1 \mp \rho)/4$ and the clearing $4\,\mathrm{den}^2(B^2 - 4C) = s_{1n}(s_{1n} \mp z) = \gamma_\pm$ (solution bijection $v = 2\,\mathrm{den}\, d$) convert the splitting of the quadratic cofactor into "$\gamma_\pm$ is a rational square" — the fibre of $C_\pm$. Signs, denominators and the direction of clearing are all carried inside the kernel statements; the same identities are certified independently in CAS [C: `terrain_crosscheck`, seven checks covering the critical points, the factorizations, the discriminant identities and the clearing].

6.4. **Integrality of the splitting surface.**

Proposition 6.3 (Integrality)**.** R_q*,* S_q *and* $R_q S_q$ *are nonsquares in* $\mathbb{Q}(a, b)$*, and remain nonsquares in* $\overline{\mathbb{Q}}(a, b)$*. Consequently* $[K(S) : K(B)] = 4$*, the* *spanning set $\{1, z, w, zw\}$ is a basis, and the splitting surface S_U is geometrically integral, and its induced dominant map to B has generic degree four.*

Over $\mathbb{Q}$ the three exclusions are kernel theorems (`Rq_not_square`, `Sq_not_square`, `RqSq_not_square`), by value specialisation at witness points ($R_q(2,3) = 2480$, $S_q(1,2) = 876$, $(R_qS_q)(1,2) = 171696$, none a rational square) [K]. A remark on method: on the chart, R_q and S_q are positive semi-definite — a quartic with four real roots has non-negative derivative discriminants — so any technique that excludes squares by exhibiting a *negative value* is unavailable for them in principle, and value specialisation at non-square values was the route adopted here. Over $\overline{\mathbb{Q}}$: each of the three has a nonconstant irreducible factor of odd multiplicity (R_q irreducible; $S_q = 3 \cdot (\text{irreducible})$; the two are coprime) [C], and a squarefree-base-change lemma [M] — an irreducible factor of odd multiplicity survives every algebraic extension of the constant field, because squarefreeness is detected by a gcd computation unchanged by extension, distinct factors share no geometric component, and a square has even valuation along every divisor; an infinite extension reduces to a finite subextension containing the finitely many coefficients of a hypothetical square root — shows none is a square in $\overline{\mathbb{Q}}(a, b)$. Degree four, the basis, and integrality follow from the standard biquadratic criterion [M]: for r, s, rs all nonsquares in a field of characteristic $\neq 2$, the algebra $K[z, w]/(z^2 - r, w^2 - s)$ is a field of degree four with basis $\{1, z, w, zw\}$. Scheme-theoretically, the coordinate ring of S_U is a free $\mathcal{O}(U_B)$-module on that basis, hence torsion-free, hence embeds in its generic localisation — the degree-four field just constructed — and a ring embedding into a field is a domain; the same chain over $\overline{\mathbb{Q}}$ gives geometric integrality [M].

6.5. **Cover nontriviality.** A square in the function field $\mathbb{Q}(a, b)$ is non-negative at every rational point where it is defined. Each of the eight candidates whose squareness would trivialise a double cover is *negative* at an explicit rational point — the eight negativity certificates are kernel lemmas at the points $(-51/13, -17/7)$ (a K-witness from Theorem 2.2) and $(1, 3)$ (`Q4_neg` through `n2Q4RqSq_neg`) — and the deduction itself is in the kernel: the specialisation principle is evaluation along a ring homomorphism (`sq_eval_nonneg`), squares descend from the fraction field to the integrally closed ring $\mathbb{Q}[a][b]$ (`square_descent`), and all eight non-squareness verdicts in $\mathbb{Q}(a, b)$ are kernel theorems (`Q4_not_square` and its seven companions). The two-system record remains as the discovery artifact [C].

For the two abstract kernel statements below, let M be any field containing $\mathbb{Q}(a, b)$ via an embedding $\iota \colon \mathbb{Q}(a, b) \hookrightarrow M$, and let $\varrho, \varsigma \in M$ satisfy $\varrho^2 = \iota(R_q)$ and $\varsigma^2 = \iota(S_q)$ (written ϱ, ς to keep them distinct from the chart quantity $\rho = z/\text{den}$); all spans are over $F := \iota(\mathbb{Q}(a, b))$, with which we identify $\mathbb{Q}(a, b)$ below — in the kernel statements the four coefficients are quantified over the base field.

Theorem 6.4 (Non-squareness in the multiquadratic extension; `Q4_not_square_multiquadratic`; `n2Q4_not_square_multiquadratic`). *In M as above, neither Q_4 nor $-2Q_4$ is the square of any element of* $\mathrm{Span}_F\{1, \varrho, \varsigma, \varrho\varsigma\}$.

The engine is a biquadratic square-class transfer (`biquadratic_transfer`): if an element of the F-span squares to $x \in F$, then one of x, xR_q, xS_q, xR_qS_q is a square in F; degenerate cases are not excluded — the case analysis absorbs them. The transfer reduces the theorem to the eight base-field exclusions above. Since every element of $K(S)$ lies in the span [M], the theorem excludes squares in all of $K(S)$; the kernel boundary is exactly the span statement, the spanning fact is ordinary mathematics.

Theorem 6.5 ($C_\pm$-exclusion in the layer; `gamma_not_square_multiquadratic`). *For either sign, $\gamma_\pm = s_{1n}(s_{1n} \mp \varrho)$ — an element of the layer $L := F(\varrho)$, not of the base field — is not, in M as above, the square of any element of* $\mathrm{Span}_F\{1, \varrho, \varsigma, \varrho\varsigma\}$.

Because $\gamma_\pm$ lives in the layer, the biquadratic transfer does not apply to it directly. Instead a layer quadratic step (`step_over_layer`) reduces a hypothetical square root to a square in the layer or its S_q-multiple, and the coordinate norm passage (`layer_square_norm`: $g_0^2 - g_1^2 R_q = (p^2 - q^2 R_q)^2$, computed in coordinates, with no Galois action) turns either case, via (12) in its polynomial form (`I2_R2`), into the statement that $-2Q_4$ is a square in $\mathbb{Q}(a, b)$ — contradicting the eight verdicts. Both signs are handled at once.

Geometric integrality of the three covers follows [M with C and K inputs]: $Q_4 = F_1F_2F_3$ with distinct irreducible factors, each to the first power, and $\gcd(F_i, R_qS_q) = 1$ [C], so Q_4 has odd valuation along every geometric component of each $F_i = 0$; the generic point of every such component lies outside the branch divisor of $S \to B$ (equivalently, the two divisors share no codimension-one component), and the valuations stay odd in $\overline{\mathbb{Q}}(S)$; hence $Q_4 \notin \overline{\mathbb{Q}}(S)^{\times 2}$, and the same divisor argument applies verbatim to $-2Q_4$. For $C_\pm$ the specialised kernel theorem quantifies its coefficients over $\mathbb{Q}(a, b)$ and is *not* base-changed; instead the abstract kernel lemmas (`step_over_layer`, `layer_square_norm` — stated for an arbitrary characteristic-zero field) are applied afresh with $K = \overline{\mathbb{Q}}(a, b)$, their inputs supplied by the odd-multiplicity divisor data through the base-change lemma, and the assembly reduces geometric squareness of $\gamma_\pm$ to that of $-2Q_4$, excluded above [M]. So C_0, C_+ and C_- are dominant, geometrically integral degree-two covers of U; removing the branch locus yields étaleness, not integrality. ($-2Q_4$ is not itself a radicand: it enters as the layer-norm obstruction class of $\gamma_\pm$.)

6.6. **What the terrain shows — and what is not claimed.** The locus inside S where Theorem 7's conclusion is algebraically forced is the proper curve $S \cap \{Q_4 = 0\}$: six one-dimensional components, each of genus 0 [C]. On each component R_q restricts to a square while S_q does not; the restriction of S_q has constant square class $12 = 4 \cdot 3$ [C], so $w^2 = S_q$ has rational solutions only where $S_q = 0$, and every rational zero of S_q on each component is a degenerate point of the chart (on F_1 there are no rational zeros at all) [C], hence inadmissible. Hence over $\mathbb{Q}$ the conclusion is never algebraically forced: *for Theorem 7's conclusion to hold at a nondegenerate* $P \in S(\mathbb{Q})$, P must lie in the image of at least one of the three nontrivial double covers — whether every terrain point does so is precisely the open content of the theorem, and is not assumed here. Thus the algebraically forced locus $S \cap \{Q_4 = 0\}$ contributes no admissible rational terrain point [C+M].

Away from the degeneracy locus each disjunct demands lifting through a double cover that is nontrivial over the splitting extension $K(S)$, so the image of the rational points of each cover is a thin subset of type II [M] — essentially by definition, as the image of the rational points of a dominant degree-two cover. *No assertion is made that* $S(\mathbb{Q})$ *contains rational points outside the union of these images*: no such conclusion follows from thinness alone, and the additional input that would provide it — for example an appropriate Hilbert-property statement for S — is not proved here; what is proved is geometric integrality. Thinness bounds what a generic-point argument can see, not what rational points exist. Within the approaches surveyed we found no finite descent reduction, and make no claim that none exists; and the field-dependence begins already at Buchholz–MacDougall's own excluded case, over the quadratic extension suggested by the square class, $\mathbb{Q}(\sqrt{3})$ $(12 = 4 \cdot 3)$, while Stroeker's points live over $\mathbb{Q}(\sqrt{3441})$.

The reading is an assessment [A] anchored by machine artifacts, not a theorem: the computations above eliminate the explicitly listed algebraically forced and cover-triviality mechanisms — they do not address whether every rational terrain point lifts to one of the covers — and what remains open is that $S(\mathbb{Q})$ has no nondegenerate points — equivalent to Conjecture 1 by Proposition 6.1. Formally, the sufficiency of Conjecture 1 (`thm7_of_conjecture1`) and the normalisation bridge are both kernel theorems; necessity of Conjecture 1 for Theorem 7, and the nonexistence of routes outside this survey, are not claimed. We make no assertion that the union of the cover images fails to contain $S(\mathbb{Q})$; no conclusion about Conjecture 1 follows from the covers alone. Every route to Theorem 7 in the survey above passes through that open statement.

7. The formalization

7.1. **Architecture and division of labour.** The development was produced under a fixed protocol separating three roles. *Statements and strategy* were designed, checked, and frozen by the author before any proof search: every definition (the gates as derivative discriminants, the curve as a Weierstrass equation, the a-map, the chart quantities, the anchor values) was cross-checked in Sage and Magma against independent reimplementations, and the frozen statement files were hashed. *Tactic-level proof filling* was then delegated to large-language-model assistants (Anthropic Claude, including its coding agent), which also drafted computer-algebra scripts and portions of this manuscript. *Verification* was never delegated: every Lean proof was re-verified by an independent run of the Lean kernel, including its per-declaration axiom report, in a process separate from the assistant runs that produced the proofs, and again by continuous integration on independent hardware; all load-bearing computer-algebra identities and numerical curve data were checked in both Magma and Sage (with PARI and the LMFDB as additional cross-checks where applicable); and no AI-generated mathematical claim was accepted without independent human checking and, where applicable, kernel or computer-algebra verification. The author selected, checked, and approved every final definition and theorem statement, and takes full responsibility for the content of this article. Under this protocol, every Lean-labelled claim below is certified by the pinned kernel; every load-bearing computer-algebra claim was reproduced independently in Magma and Sage; ordinary mathematical deductions and literature claims ([M], [A]) were checked by the author and lie outside the kernel/computer-algebra trust boundary — and nothing below rests on unverified model output.

7.2. **The artifact: pins, certificates, and continuous verification.** The development compiles against mathlib (toolchain `v4.31.0-rc1`, mathlib commit `d568c8c`) and is archived at the frozen release v1.1.0 (commit `a5cf446`) of `https://github.com/hirokifukui/rd-quartics-classification`, with concept DOI `10.5281/zenodo.21766032`. Every declaration cited in this paper is hyperlinked to its file and line *at that tag*, so the links are permanent; the ten pins listed in the repository README are re-resolved mechanically against the actual tag content on every push (a fail-closed check: a pin that no longer matches fails the build). The dependency graph, with one node per formal declaration tracked in the blueprint and kernel status marked per node, is at `https://hirokifukui.github.io/rd-quartics-classification/blueprint/`.

All computer-algebra-derived certificates — the `linear_combination` cofactors of §§4.2–4.3, the elimination certificates, and the Nullstellensatz cofactors of the sign cover — are reproduced *verbatim* in the Lean source, so their correctness is checked by the kernel rather than trusted; the generating scripts remain in `scripts/` as the discovery record. Continuous integration runs three independent jobs on every push: the Lean build with a named-declaration check (all 137 blueprint declarations must exist) and an axiom assertion for the main theorems; a computer-algebra re-execution job that reruns the terrain cross-check script in a digest-pinned container and diffs its 42 checks against the committed log; and a tag self-verification job that rebuilds the release tag itself.

7.3. **The axiom footprint, and the history it records.** The kernel reports the axiom footprint of `thm7prime`, of the transcript theorems of Part I, and of the boundary theorems of Part III as exactly `[propext,` `Classical.choice, Quot.sound]` — the standard axioms of classical mathlib reasoning, with no custom axiom and no `sorry`. The development retains, deliberately, one disclosed axiom (`rank_E576i2`: the Mordell–Weil rank of `576i2` is 1, verified unconditionally in two independent systems) together with the auxiliary declarations that consume it. These are the trace of an earlier *conditional* architecture: the forward direction of Theorem 7′ was first proved under the rank hypothesis, and the unconditional reconstruction of §4.3 superseded it. The axiom was not deleted, because the kernel's per-declaration axiom report then documents the history of the proof as well as its final status: a reader can verify, from the report alone, both that the main theorem is unconditional and that it was not always so. Mathlib's lack of a Mordell–Weil rank theory (§8) is thus visible in the artifact twice — once as the wall that stands for Conjecture 1, and once as the pressure that produced the rank-free proof; the disclosed axiom marks the exact point where the pressure was released. An archived copy of the pre-merge master file, carrying the same axiom, is retained under `archive/` as provenance and is excluded from the live build.

7.4. **Quantitative summary.**

component	files	lines	theorems	definitions	axioms
`Gap/` (audit + boundary)	9	3,271	148	53	0
`Thm7Prime/` (repair)	3	1,076	61	27	1
live development	12	4,347	209	80	1
`archive/` (provenance)	3	783	40	30	(1)

Declaration counts are top-level `theorem` and `def`-family declarations at the frozen tag; the parenthesised axiom is the archived copy of the same `rank_E576i2`. The named-declaration check covers 137 blueprint declarations. The blueprint itself runs to 1,378 lines of LaTeX against 4,347 lines of live Lean, a ratio of roughly 1 : 3.2. The verification scripts comprise 29 files across four suites (Magma, Sage, rank, terrain), with a certificate index in `scripts/CERTIFICATES.md`. With the pinned mathlib pre-built, a full kernel verification of the development (all twelve files, `lake build`) completes in 35 seconds of wall time (88 seconds of CPU time) on an Apple M4 Pro with 48 GB of memory; re-checking the statement file `Thm7Prime/Master.lean` alone takes 10 seconds. This manuscript runs to 2,052 lines of LaTeX against the same 4,347 lines of Lean, a ratio of roughly 1 : 2.1.

7.5. **Design decisions.** Three decisions shaped the development and are recorded for reuse. First, the *gate form*: rational-derivedness was formalized through division-free discriminant gates rather than through `Polynomial.roots` directly, because polynomial identities under `linear_combination` are where the kernel is strong; the cost — a possible statement-level slip between the gate form and the natural form — was then paid down by proving the equivalence in the kernel (Theorem 5.1). Second, *value specialisation over sign* *arguments*: since R_q and S_q are positive semi-definite on the chart (§6.4), negativity certificates were unavailable for them in principle, and non-square *values* at rational points carried the exclusions instead; the sign-based route and the value-based route coexist in the eight verdicts, each where it applies. Third, *abstract lemmas over base change*: the layer lemmas of §6.5 were stated over an arbitrary characteristic-zero field from the outset, so that the geometric ($\overline{\mathbb{Q}}$) applications are fresh instantiations of kernel theorems rather than informal base-change arguments — the kernel boundary stays exactly where the paper says it is.

8. The open sector

Following the practice of stating unproved propositions as formal definitions — as [11] do for the Riemann hypothesis — the outstanding statement is displayed here in the exact form it takes in the development (`Conjecture1_normal`; reproduced verbatim, deliberately without proof and without `sorry` — introduced as a definition of a proposition rather than as a theorem: the development contains no proof term inhabiting it, so the source itself keeps the syntactic distinction between stating the conjecture and proving it):

```
def Conjecture1_normal (K : Type*) [Field K] : Prop :=
  ∀ a b : K, DistinctRoots a b → ¬ KDerived (bmQuartic a b)
```

One kernel theorem hangs from it (`thm7_of_conjecture1`): if Conjecture 1 (normalised) holds, then the promised conclusion T0 of §2.2.4 holds — vacuously, since the conjecture empties the hypothesis class. This one line is the only route to T0 represented in the formal dependency graph; what it proves is sufficiency, not necessity. By the normalisation bridge (Proposition 6.1), Conjecture 1 is equivalent to the nonexistence of nondegenerate rational points on the splitting surface S; Part III records how far computation shuts the known approaches, and what a proof would have to decide.

One standing obstacle is a property of the current library, not of the mathematics: at the pinned commit, mathlib does not provide the machinery needed here for Mordell–Weil rank, Selmer computations, Chabauty's method, or Coleman integration, so any completion of the conjecture that proceeds through the arithmetic of elliptic surfaces cannot at present be carried to a kernel-checked proof without either substantial new formalization or disclosed axioms encapsulating the computer-algebra rank computations. For the repeated-root sector this wall was *bypassed* rather than breached — Theorem 7′ is rank-free — and it remains standing exactly for Conjecture 1. The reopening condition is explicit: the required machinery must be formalised, either in mathlib or locally. Dated computational records (measured fibres, descent runs in progress, the survey of cover mechanisms) are kept as research notes at the repository root, separated from the durable blueprint.

Conjecture 1 itself remains open — this development maps its boundary, it does not resolve it.

Appendix A. Explicit data and the formal statement

The sign-selection sextic of §4.3(c) is

$$\begin{aligned}P(a,r,s) = {} & -15064695\,ars^4 + 9029055\,as^5 - 2983275\,rs^5 \\ & + 4985475\,s^6 + 106313580\,ars^3 - 112603320\,as^4 \\ & + 35756697\,rs^4 - 42229133\,s^5 - 143856972\,ars^2 \\ & + 415929600\,as^3 - 121880772\,rs^3 + 95245548\,s^4 \\ & - 394608672\,ars - 21311424\,as^2 - 48676140\,rs^2 \\ & + 103362048\,s^3 + 833400576\,ar - 2611061568\,as \\ & + 934428960\,rs - 758843424\,s^2 + 3732293376\,a \\ & - 1245777408\,r + 1320338880\,s - 1009822464.\end{aligned}$$

The remaining explicit data (the `linear_combination` cofactors, the elimination certificates, and the six Nullstellensatz cofactors) are reproduced in the Lean source and checked by the kernel; the main text records the identity each certificate proves, and the complete coefficient lists live in the archived development.

Following the journal's guidance that in-text code should complement rather than reproduce the artifact, we display only the main statement; its supporting definitions are quoted and pinned at their points of use in §4.1, and the full statement file is `Thm7Prime/Master.lean`. The statements below are reproduced verbatim from that file at the frozen tag; the correspondence between them and Theorem 4.1 is the statement-fidelity claim of this paper, checkable by inspection together with Theorem 5.1.

```
def Thm7Backward : Prop :=
  ∀ w z : ℚ, OnE w z → aDen w z ≠ 0 → aMap w z ≠ 0 → aMap w z ≠ 1 →
    RD211 (aMap w z)

def Thm7Forward : Prop :=
  ∀ a : ℚ, RD211 a → ∃ w z : ℚ, OnE w z ∧ aDen w z ≠ 0 ∧ aMap w z = a

def Thm7Prime : Prop := Thm7Backward ∧ Thm7Forward

theorem thm7prime : Thm7Prime := ⟨thm7_backward, thm7_forward⟩
```

Code availability

The Lean development, all Magma and Sage verification scripts and logs, the blueprint, and build instructions are maintained at `https://github.com/hirokifukui/rd-quartics-classification` and archived at the frozen release v1.1.0 (commit `a5cf446`); the release is deposited at Zenodo under concept DOI `https://doi.org/10.5281/zenodo.`

`21766032`. The development descends from two earlier archived artifacts, from which selected scripts were carried over with a transfer manifest: the audit repository (`https://doi.org/10.5281/zenodo.21465598`) and the repair repository (`https://doi.org/10.5281/zenodo.21470001`). The release revision carries the intrinsic Software Heritage identifier `swh:1:rev:a5cf446d09b148c9c27a5232e0a0bfddd1f689ab`.

Acknowledgment

We record with sadness that Professor R. J. Stroeker died in October 2025, and Professor J. A. MacDougall in May 2025, while the verification reported here was in preparation. The corrections reported here are offered to the record of bodies of work that our audit found overwhelmingly sound; we have therefore taken particular care to distinguish typographical errors, valid intermediate results, and the inferential gap, and to state which substantive conclusions of the cited works remain unaffected.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares no competing interests.

References

[1] R. H. Buchholz, S. M. Kelly, On rational-derived quartics, Bull. Austral. Math. Soc. 51 (1995) 121–132. `https://doi.org/10.1017/S0004972700013940`

[2] R. H. Buchholz, J. A. MacDougall, When Newton met Diophantus: a study of rational-derived polynomials and their extension to quadratic fields, J. Number Theory 81 (2000) 210–233. `https://doi.org/10.1006/jnth.1999.2473`

[3] A. Bremner, B. Carrillo, On K-derived quartics, J. Number Theory 168 (2016) 276–291. `https://doi.org/10.1016/j.jnt.2016.04.024`

[4] A. J. Best, C. Birkbeck, R. Brasca, E. Rodriguez Boidi, R. Van de Velde, A. Yang, A complete formalization of Fermat's Last Theorem for regular primes in Lean, Ann. Formaliz. Math. 1 (2025) 103–132. `https://doi.org/10.46298/afm.14586`

[5] B. Carrillo, Rational Derived Quartics and Related Topics, Ph.D. thesis, Arizona State University, 2019.

[6] C. E. Carroll, Polynomials all of whose derivatives have integer roots, Amer. Math. Monthly 96 (1989) 129–130.

[7] J. E. Cremona, Algorithms for Modular Elliptic Curves, second ed., Cambridge University Press, Cambridge, 1997.

[8] E. V. Flynn, On $\mathbb{Q}$-derived polynomials, Proc. Edinb. Math. Soc. (2) 44 (2001) 103–110. `https://doi.org/10.1017/S0013091599000760`

[9] R. K. Guy, Unsolved problems come of age, Amer. Math. Monthly 96, no. 10 (1989) 903–909.

[10] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, in: Automated Deduction — CADE 28, LNCS 12699, Springer, 2021, pp. 625–635.

[11] D. Loeffler, M. Stoll, Formalizing zeta and L-functions in Lean, Ann. Formaliz. Math. 1 (2025) 43–56. `https://doi.org/10.46298/afm.15328`

[12] J. Ludwig, C. Merten, Formalising the Bruhat–Tits Tree, Ann. Formaliz. Math. 2 (2026) 55–81. https://doi.org/10.46298/afm.15738

[13] The mathlib Community, The Lean mathematical library, in: Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), pp. 367–381. https://doi.org/10.1145/3372885.3373824

[14] R. J. Stroeker, On $\mathbb{Q}$-derived polynomials, Econometric Institute Report EI 2002-30, Erasmus University Rotterdam, 2002.

[15] R. J. Stroeker, On $\mathbb{Q}$-derived polynomials, Rocky Mountain J. Math. 36 (2006) 1705–1713.

Research Institute of Criminal Psychiatry / Sex Offender Medical Center, Tokyo, Japan; and Department of Neuropsychiatry, Kyoto University, Kyoto, Japan. ORCID: 0009-0008-7122-522X

Email address: fukui@somec.org